Molinda Mihiranga

14/A/1, Aluth Ela Road, Badulla | +94 743872722 | molindamihiranga17@gmail.com | molindamihiranga17.github.io | linkedin.com/in/molinda-mihiranga | github.com/MolindaMihiranga17

EDUCATION

Sri Lanka Institute of Information Technology (SLIIT) Malabe, Sri Lanka

Bachelor of Science in Information Technology (Specialized in IT) 2022 – present

Badulla Central College Badulla, Sri Lanka

Science Stream (Mathematics) 2013 – 2021

Projects

Vettly: Smart Recruitment Platform with AI Screening GitHub Live Website

ASP.NET Core, Go, Python FastAPI, React, TypeScript, PostgreSQL, Redis, SignalR, Cloudflare R2, Docker

- Architected a polyglot microservices monorepo — seven ASP.NET Core services, two Python/FastAPI AI services, and two Go services — sharing one PostgreSQL instance and Redis for caching and async events.
- Built the Go api-gateway as the platform's single public entry point, handling JWT validation, rate limiting, CORS, and reverse proxying (with WebSocket passthrough) to every backend service.
- Developed the recruiter web dashboard (React 19 + TypeScript, TanStack Query, Zustand) with a drag-and-drop Kanban pipeline, candidate deep-dive pages with AI/match scores and skill-gap breakdowns, and a bias detection UI.
- Built matching-service, a semantic candidate ranking engine using sentence-transformer embeddings, and an async screening-service that scores resumes for AI fit and bias flags.
- Implemented real-time messaging (SignalR) and an e-signature offer flow with PDF generation, Cloudflare R2 storage, and a SHA-256-hashed audit trail; deployed the full stack via GitHub Actions to free-tier hosting.
- Deployed the live platform with the React frontend on Cloudflare Pages and 11 containerized backend services on Render, using Neon PostgreSQL, Upstash Redis, Cloudflare R2 object storage, and GitHub Container Registry; GitHub Actions uses path-based change detection to build and redeploy only modified services.

HydroGuardian: Smart Hydroponic System GitHub

React Native, Expo, FastAPI, TensorFlow, CNN+LSTM, ESP32, WebSockets, PyJWT

- Built an AI-driven automated hydroponic fodder system for Sri Lankan farmers, with ESP32 IoT integration for real-time sensor monitoring (temperature, humidity, pH, EC, TDS) and actuator control over a WebSocket gateway.
- Developed a full-stack Expo React Native mobile app (15+ screens) covering onboarding, a setup wizard, live monitoring dashboards, and energy tracking, built from 8 reusable UI components.
- Built the FastAPI backend with JWT authentication, Swagger/Redoc API docs, and a WebSocket gateway for ESP32 device discovery, health monitoring, and real-time data broadcasting, including ESP32-CAM streaming.
- Implemented multi-modal plant disease detection (CNN+LSTM), seed quality identification, and RL-based energy consumption optimization.

SpendWise: Personal Finance & Vehicle Tracker GitHub

React Native, Expo, Go, Spring Boot, Python FastAPI, Apache Kafka, PostgreSQL, Redis, Kong, AWS

- Designed an event-driven microservices backend (Auth in Go, Finance in Spring Boot, Vehicle in Go/Gin, Notifications+AI in FastAPI) behind a Kong API gateway, with all inter-service communication over Apache Kafka topics.
- Built a dedicated vehicle expense module — fuel log with auto-calculated efficiency and cost-per-km, maintenance history, and Sri Lanka-specific reminders — publishing events consumed by the Finance service so costs post automatically with no double-entry.
- Implemented AI-powered spending insights and fuel anomaly diagnostics via the Anthropic Claude API, streamed to the mobile app in real time over SSE.

- Built the reminder and push notification engine (APScheduler, Expo Push API) with deep-linking, and designed the offline-first Expo app (Expo SQLite, Zustand + TanStack Query) with a freemium paywall.

SKILLS

Languages:	Java, C, C++, JavaScript, TypeScript, Go, Python, PHP, Kotlin, SQL
Web & Mobile:	React.js, React Native, Next.js, Express.js, Node.js, ASP.NET Core, Spring Boot, FastAPI
Database & Messaging:	PostgreSQL, MongoDB, MySQL, Redis, Apache Kafka
Cloud & Infra:	Docker, Kong, Cloudflare R2, AWS, GitHub Actions
Developer Tools:	Git, R Studio, PowerBI, Trello, Jira, Playwright

REFERENCES

Mr. Uditha Senavirathna

+94 77 761 5403
Senior Tech Lead
Intervest Software Technologies (Pvt) Ltd

Mrs. Arundika Gunarathne

Head of Department of Marketing Management
+94 775176904
arundika@kiu.ac.lk
Faculty of Management, KAATSU International University